

Using range() to Make a List of Numbers

If you want to make a list of numbers, you can convert the results of `range()` directly into a list using the `list()` function. When you wrap `list()` around a call to the `range()` function, the output will be a list of numbers.

In the example in the previous section, we simply printed out a series of numbers. We can use `list()` to convert that same set of numbers into a list:

```
numbers = list(range(1, 6))
print(numbers)
```

This is the result:

```
[1, 2, 3, 4, 5]
```

We can also use the `range()` function to tell Python to skip numbers in a given range. If you pass a third argument to `range()`, Python uses that value as a step size when generating numbers.

For example, here's how to list the even numbers between 1 and 10:

```
even_numbers.py numbers = list(range(2, 11, 2))
print(even_numbers)
```

In this example, the `range()` function starts with the value 2 and then adds 2 to that value. It adds 2 repeatedly until it reaches or passes the end value, 11, and produces this result:

```
[2, 4, 6, 8, 10]
```

You can create almost any set of numbers you want to using the `range()` function. For example, consider how you might make a list of the first 10 square numbers (that is, the square of each integer from 1 through 10). In Python, two asterisks (`**`) represent exponents. Here's how you might put the first 10 square numbers into a list:

```
square
_numbers.py squares = []
for value in range(1, 11):
❶   square = value ** 2
❷   squares.append(square)

print(squares)
```

We start with an empty list called `squares`. Then, we tell Python to loop through each value from 1 to 10 using the `range()` function. Inside the loop, the current value is raised to the second power and assigned to the variable `square` ❶. Each new value of `square` is then appended to the list `squares` ❷. Finally, when the loop has finished running, the list of squares is printed:

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```
